Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Consider a family of circles which are passing through the point $(-1, 1)$ and are tangent to x-axis. If (h, k) are the coordinates of the centre of the circles, then the set of values of k is given by the interval

(A) $0 < k < 1/2$

(B) $k \geq 1/2$

(C) $-1/2 \leq k \leq 1/2$

(D) $k \leq 1/2$

Q.2

Statement-1: Limiting points of a family of coaxial circles are $(1, 2)$ and $(2, 1)$. No circle of this family passes through the origin.
Statement-2 : Equation of a circle passing through $(1, 2)$ and $(2, 1)$, the centre of which does not lie on the join of these points is $x^2 + y^2 - 3x - 3y + 4 = 0$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.3

Statement-1: Limiting points of a family of co-axial system of circles are $(1, 1)$ and $(3, 3)$. The member of this family passing through the origin is $2x^2 + 2y^2 - 3x - 3y = 0$
Statement-2: Equation of the tangent to the circle $2x^2 + 2y^2 - 3x - 3y = 0$ at the origin is $x + y = 0$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

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Q.4

Equation of a family of circles, such that each member of the family passes through the origin and makes an intercept on the line $y = 2x$ which is twice the intercept made on the line $x = 2y$ is (λ being a parameter)

(A) $x^2 + y^2 - 2\lambda y = 0$

(B) $x^2 + y^2 - 2\lambda x = 0$

(C) $x^2 + y^2 - 2\lambda x - 2\lambda y = 0$

(D) none of these

Q.5

Equation of a common chord of the circles $x^2 + y^2 + 6x - 10y + 9 = 0$ and $x^2 + y^2 - 10x + 6y + 25 = 0$ is

(A) $x + y + 4 = 0$

(B) $x - y + 4 = 0$

(C) $x + y + 1 = 0$

(D) $x - y - 1 = 0$

Q.6

An equilateral triangle is inscribed in the circle $x^2 + y^2 = a^2$ with the vertex at $(a, 0)$. The equation of the side opposite to this vertex is

(A) $2x - a = 0$

(B) $x + a = 0$

(C) $2x + a = 0$

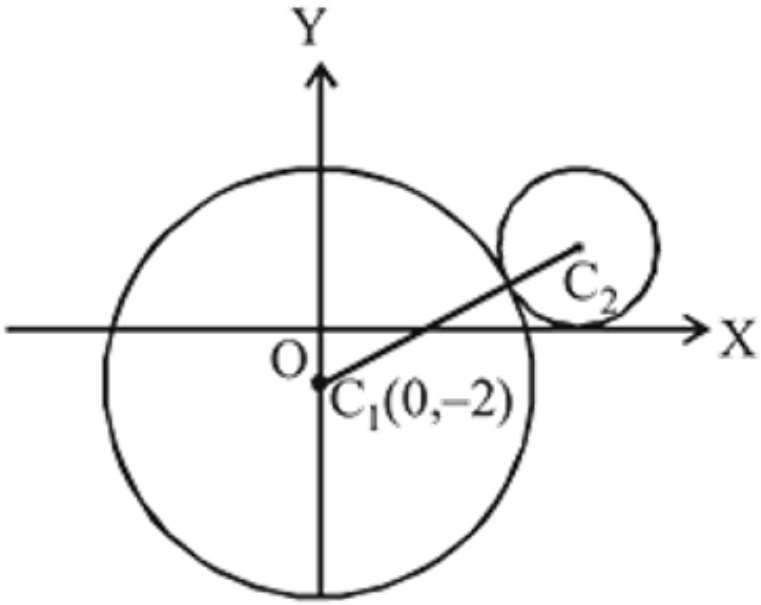
(D) $3x - 2a = 0$

Q.7

In the given figure, the equation of the larger circle is $x^2 + y^2 + 4y - 5 = 0$ and the distance between

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centres is 4. Then the equation of smaller circle is



(A) $(x-\sqrt{7})^2+(y-1)^2=1$

(B) $(x+\sqrt{7})^2+(y-1)^2=1$

(C) $x^2+y^2=2\sqrt{7}x+2y$

(D) None of these

Q.8

Distances from the origin to the centres of the three circles $x^2+y^2-2\lambda_i x=c^2$ (where c is constant and $\mathrm{i}=1,2,3$) are in G.P. Then the lengths of tangents drawn from any point on the circle $x^2+y^2=c^2$ to these circles are in

(A) A.P.

(B) GP.

(C) H.P.

(D) None

Q.9

The common chord of $x^2+y^2-4x-4y=0$ and $\mathrm{x}^2+\mathrm{y}^2=16$ subtends at the origin an angle equal to

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

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Q.10

If the line $x\cos\alpha + y\sin\alpha = p$ represents the common chord of the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 + b^2(a > b)$, where A and B lie ont he first circle and P and Q lie on the second circle, then AP is equal to

(A) $\sqrt{\mathrm{a}^2 + \mathrm{p}^2} + \sqrt{\mathrm{b}^2 + \mathrm{p}^2}$

(B) $\sqrt{\mathrm{a}^2 - \mathrm{p}^2} + \sqrt{\mathrm{b}^2 - \mathrm{p}^2}$

(C) $\sqrt{\mathrm{a}^2 - \mathrm{p}^2} - \sqrt{\mathrm{b}^2 - \mathrm{p}^2}$

(D) $\sqrt{\mathrm{a}^2 + \mathrm{p}^2} - \sqrt{\mathrm{b}^2 + \mathrm{p}^2}$

Q.11

The length of the chord $x + y = 3$ intercepted by the circle $x^2 + y^2 - 2x - 2y - 2 = 0$ is

(A) $\frac{7}{2}$

(B) $\frac{3\sqrt{3}}{2}$

(C) $\sqrt{14}$

(D) $\frac{\sqrt{7}}{2}$

Q.12

Consider a circle of radius R. What is the length of a chord which subtends an angle θ at the centre?

(A) $2\mathrm{R}\sin\left(\frac{\theta}{2}\right)$

(B) $2\mathrm{R}\sin\theta$

(C) $2\mathrm{R}\tan\left(\frac{\theta}{2}\right)$

(D) $2\mathrm{R}\tan\theta$

Q.13

From the origin, chords are drawn to the circle $(x-1)^2 + y^2 = 1$, then equation of locus of middle points of these chords, is -

(A) $x^2 + y^2 = 1$

(B) $x^2 + y^2 = x$

(C) $x^2 + y^2 = y$

(D) None of these

Q.14

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The equation of the image of circle $x^2+y^2+16x-24y+183=0$ by the line mirror $4x+7y+13=0$ is

(A) $x^2+y^2+32x-4y+235=0$

(B) $x^2+y^2+32x+4y-235=0$

(C) $x^2+y^2+32x-4y-235=0$

(D) $x^2+y^2+32x+4y+235=0$

Q.15

If from any point P, tangents PT, PT' are drawn to two given circles with centres A and B respectively; and if PN is the perpendicular from P on their radical axis, then
$\mathrm{PT}^2-\mathrm{PT}'^2=$

(A) PN.AB

(B) $2\mathrm{PN}\cdot\mathrm{AB}$

(C) $4\mathrm{PN}\cdot\mathrm{AB}$

(D) None of these

Answers & Solutions

Q.1 Answer:

$k\geq 1/2$

Solution:

Equation of the circle with centre (h,k) and touching x-axis is
$(x-h)^2+(y-k)^2=k^2$
which passes through $(-1,1)$ if $h^2+2h-2k+2=0$
or if $(h+1)^2=2k-1$
For real values of h, $2k-1\geq 0$.
$\Rightarrow k\geq 1/2$

Q.2 Answer:

STATEMENT-1 is True, STATEMENT-2 is False

Solution:

In statement 1, two members of the family are $(x-1)^2+(y-2)^2=0$ and $(x-2)^2+(y-1)^2=0$. Equation of any member of this family is $x^2+y^2-2x-4y+5+\lambda\left(x^2+y^2-4x-2y+5\right)=0$ which passes through the origin if $\lambda=-1$, in which case the equation does not represent a circle. So the statement is true.

In Statement-2, Equation represents the circle on the join of the given points as diameter. So the statement is false.

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Q.3 Answer:

STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

Solution:

In statement-1, two members of the co-axial system are circles with centres at $(1,1),(3,3)$ and radius zero. So the equation of the system of circles is $(x-1)^2+(y-1)^2+\lambda\left[(x-3)^2+(y-3)^2\right]=0$

If it passes through the origin. $1+1+\lambda[9+9]=0 \Rightarrow \lambda=-1/9$ and the required circle is $2x^2+2y^2-3x-3y=0$

So statement- 1 is true. Statement-2 is also true but does not lead the statement-1.

Q.4 Answer:

$x^2+y^2-2\lambda y=0$

Solution:

Let the equation of the required circles be

$x^2+y^2+2gx+2fy=0$

Intercept on the line $y=2x$ is $\frac{2g+4f}{5}$

Intercept on the line $x=2y$ is $\frac{4g+2f}{5}$

$\frac{2g+4f}{5}=2\times\frac{4g+2f}{5} \Rightarrow g=0$

and the required equation is $x^2+y^2+2fy=0$ or $x^2+y^2-2\lambda y=0$

where λ is a parameter.

Q.5 Answer:

$x-y-1=0$

Solution:

Equation of the common chord is $x^2+y^2+6x-10y+9-$

$\left(x^2+y^2-10x+6y+25\right)=0$

$\Rightarrow \quad x-y-1=0$

Q.6 Answer:

$2x+a=0$

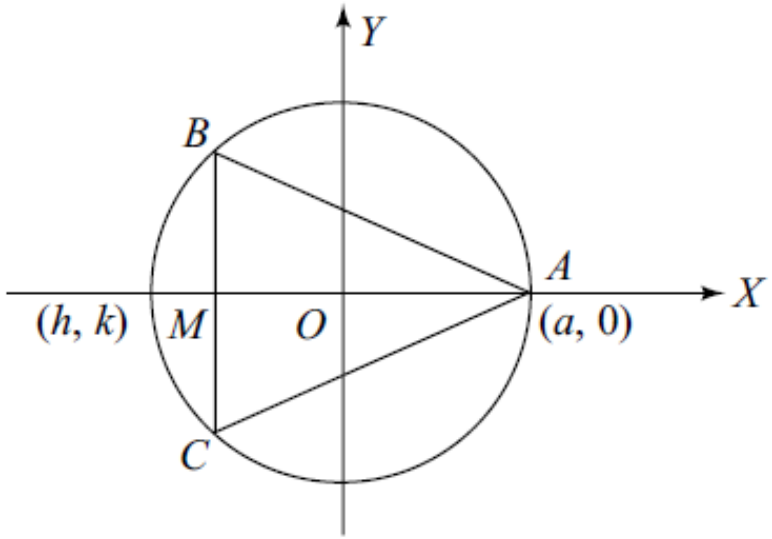
Solution:

$A(a,0)$ be the vertex of the equilateral triangles ABC inscribed in the circle $x^2+y^2=a^2$

Let M be the middle point of the side BC, then MOA is perpendicular to BC and O being the

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centroid of the triangle $OA = 2(OM)$



(Circumcentre and Centroid of an equilateral triangle are same.) So if (h, k) be the coordinates of M, then $\frac{2h+a}{3} = 0$ and $\frac{2k+0}{3} = 0$

$\Rightarrow h = -(a/2)$ and $k = 0$

and hence the equation of BC is $x = -a/2$ or $2x + a = 0$.

Q.7 Answer:

$(x - \sqrt{7})^2 + (y - 1)^2 = 1$

Solution:

We have $x^2 + y^2 + 4y - 5 = 0$. Its centre is $\mathrm{C}_1(0, -2)$

$\mathrm{r}_1 = \sqrt{4 + 5} = 3$. Let $\mathrm{C}_2(\mathrm{h}, \mathrm{k})$ be the centre of the smaller circle and its radius r_2. Then $\mathrm{C}_1\mathrm{C}_2 = 4 \Rightarrow \sqrt{h^2 + (k+2)^2} = 3 + r_2 = 4$

$\Rightarrow \mathrm{r}_2 = 1$

But $\mathrm{k} = \mathrm{r}_2 = 1$

[it touches x-axis]

- From eq $(1), 4 = \sqrt{\mathrm{h}^2 + (1+2)^2}$

$\Rightarrow 16 = \mathrm{h}^2 + 9 \Rightarrow \mathrm{h}^2 = 7 \Rightarrow \mathrm{h} = \pm\sqrt{7}$

Since $\mathrm{h} > 0 \quad \therefore \mathrm{h} = \sqrt{7}$

Hence, required circle is $(x - \sqrt{7})^2 + (y - 1)^2 = 1$

Q.8 Answer:

GP.

Solution:

Centres of the circle are $(\lambda_1, 0), (\lambda_2, 0)$ and $(\lambda_3, 0)$

$\therefore$ Given $\lambda_2^2 = \lambda_1\lambda_3$

Any point on $x^2 + y^2 = c^2$ is $(c\cos\theta, c\sin\theta)$

The length of tangents from this point to the given circles are :

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$t_1^2 = -2\lambda_1 c\cos\theta, t_2^2 = -2\lambda_2 c\cos\theta$

$t_3^2 = -2\lambda_3 c\cos\theta$

Clearly, $\left(t_2^2\right)^2 = t_1^2 \cdot t_3^2 \Rightarrow \left(\because \lambda_2^2 = \lambda_1\lambda_3\right)$

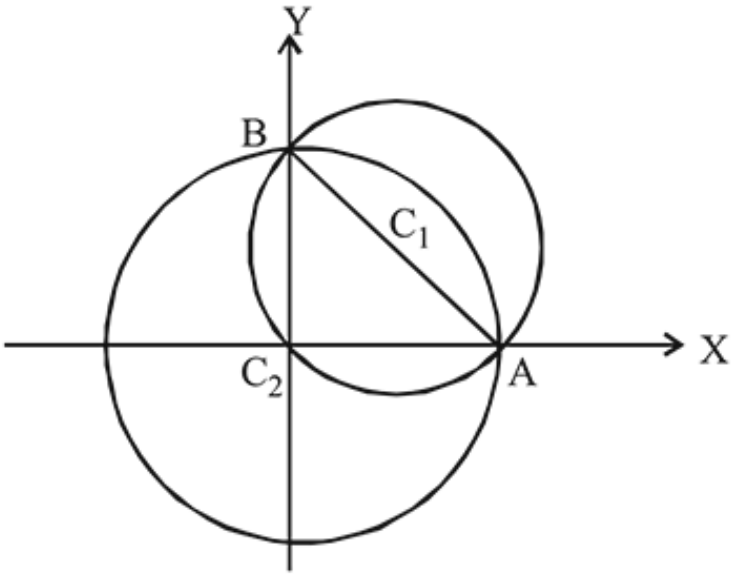
$\therefore t_1^2, t_2^2, t_3^2$ **are in G.P. hence** t_1, t_2, t_3 **are also in GP.**

Q.9 Answer:

$\frac{\pi}{2}$

Solution:

The centre of two circles are $C_1(2,2)$ **and** $C_2(0,0)$**. The radii of two circles are** $r_1 = 2\sqrt{2}$ **and** $r_2 = 4$



The eq. of the common chord of the circles $x^2 + y^2 - 4x - 4y = 0$ **and** $x^2 + y^2 = 16$ **is** $x + y = 4$

which meets the circle $x^2 + y^2 = 16$ **at points** $A(4,0)$ **and** $B(0,4)$**. Obviously** $OA \perp OB$**. Hence, the common chord** AB **makes a right angle at the centre of the circle** $x^2 + y^2 = 16$**. Where,** O **is the origin and the centre** C_2 **of the second circle.**

Q.10 Answer:

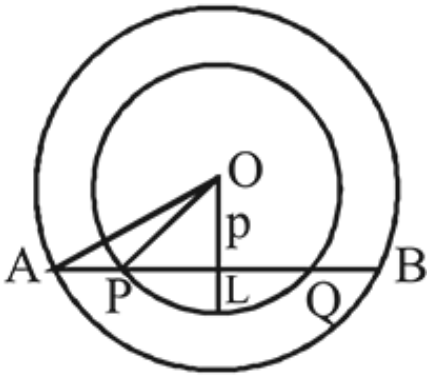
$\sqrt{a^2 - p^2} - \sqrt{b^2 - p^2}$

Solution:

The given circles are concentric with centre at $(0,0)$ **and the length of the perpendicular from** $(0,,$ **0) on the given line is**

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p. Let $\mathrm{OL} = \mathrm{p}$



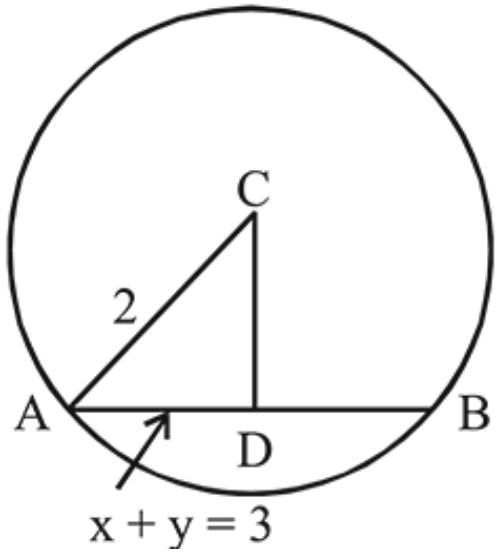
then, $AL = \sqrt{OA^2 - OL^2} = \sqrt{a^2 - p^2}$
and $PL = \sqrt{OP^2 - OL^2} = \sqrt{b^2 - p^2}$
$\Rightarrow AP = \sqrt{a^2 - p^2} - \sqrt{b^2 - p^2}$

Q.11 Answer:

$\sqrt{14}$

Solution:

The centre of the circle is $\mathrm{C}(1, 1)$ **and radius of the circle is 2, perpendicular distance from** C **on** AB**, the chord** $x + y = 3$



$\mathrm{CD} = \left|\frac{1+1-3}{\sqrt{2}}\right| = \frac{1}{\sqrt{2}}$
$\therefore \mathrm{AD} = \sqrt{4 - \frac{1}{2}} = \sqrt{\frac{7}{2}} \quad \left[\mathrm{AD} = \sqrt{\mathrm{AC}^2 - \mathrm{CD}^2}\right]$
Hence, the length of the chord $\mathrm{AB} = 2\mathrm{AD} = 2\sqrt{\frac{7}{2}} = \sqrt{14}$

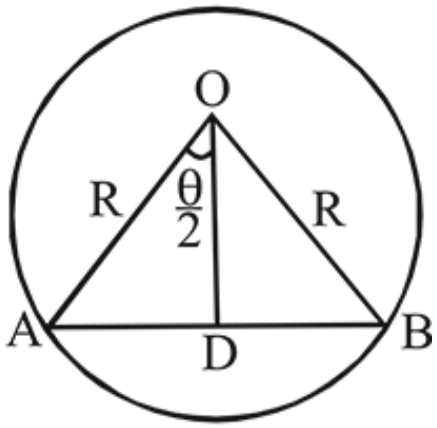
Q.12 Answer:

$2\mathrm{R}\sin\left(\frac{\theta}{2}\right)$

Solution:

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Let there be a circle of radius R and AB a chord. $\mathrm{OD} \perp \mathrm{AB}$ and $\mathrm{AD} = \mathrm{DB}$.
and $AD = 2AD \angle \mathrm{AOB} = \theta$
$\Rightarrow \quad \angle \mathrm{AOD} = \frac{\theta}{2}$



In $\triangle \mathrm{AOD}$,
$\sin \frac{\theta}{2} = \frac{\mathrm{AD}}{\mathrm{OA}}$
$\sin \frac{\theta}{2} = \frac{\mathrm{AD}}{\mathrm{R}}$
$\mathrm{AD} = \mathrm{R} \sin \frac{\theta}{2}$
$\therefore \quad$ Length of chord $\mathrm{AB} = 2\ \mathrm{AD} = 2\mathrm{R} \sin \frac{\theta}{2}$.

Q.13 Answer:

$x^2 + y^2 = x$

Solution:

Here equation of the given circle is $x^2 + y^2 - 2x = 0$ This clearly passes through origin Hence if (x_1, y_1) be midpoint of the chord then its equation is given by $T = S_1 \Rightarrow xx_1 + yy_1 - (x + x_1) = x_1^2 + y_1^2 - 2x_1$
or $xx_1 + yy_1 - x = x_1^2 + y_1^2 - x_1$
This passes through the origin $(0, 0)$
$\therefore \quad x_1^2 + y_1^2 - x_1 = 0$
$\therefore \quad$ Required locus is $x^2 + y^2 = x$

Q.14 Answer:

$x^2 + y^2 + 32x + 4y + 235 = 0$

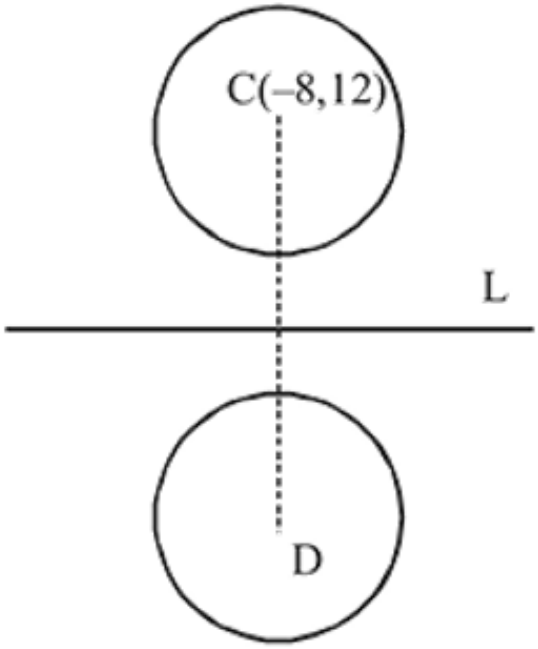
Solution:

The centre of the given circle is $(-8, 12)$ and radius is 5 .

The image of the circle will have the same radius,
i.e. the radius of the required circle is 5 . The centre D of the required circle is the image of the centre

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C of the given circle in the line mirror. If D be (α, β) then



$\frac{\alpha+8}{4} = \frac{\beta-12}{7} = -2\left[\frac{4\times-8+7\times12+13}{4^2+7^2}\right]$

[See straight line]

Or, $\frac{\alpha+8}{4} = \frac{\beta-12}{7} = \frac{-2\times65}{65} = -2$

$\therefore \alpha = -16, \beta = -2$

$\therefore$ **Required circle is** $(x+16)^2 + (y+2)^2 = 5^2$

Q.15 Answer:

$2PN \cdot AB$

Solution:

Let the two given circles be $x^2 + y^2 + 2g_1x + c = 0$

and $x^2 + y^2 + 2g_2x + c = 0$

Their centres are $A(-g_1, 0)$ **and** $B(-g_2, 0)$

$\therefore AB = g_1 - g_2$

Let P **be the point** (x_1, y_1) **. Then,** $PT = \sqrt{x_1^2 + y_1^2 + 2\,g_1x_1 + c}$

$PT = \sqrt{x_1^2 + y_1^2 + 2\,g_2x_1 + c}$

Radical axis of (1) **and** (2) **is** $2\,(\,g_1 - g_2)\,x = 0$ **or** $x = 0, PN =$ **length of** $\perp$ **from** P **on radical axis** $= x_1 \therefore PT^2 - PT'^2$

$= (x_1^2 + y_1^2 + 2\,g_1x_1 + c) - (x_1^2 + y_1^2 + 2\,g_2x_1 + c)$

$= 2x_1\,(\,g_1 - g_2) = 2PN \cdot AB$

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